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TD N°4– Reseaux

Niveau : L3

Date : Le/4/20/26

Objectif du devoir

Dans ce TD, vous devez :

Concevoir un réseau de campus universitaire

Mettre en place une topologie multi-routeurs

Organiser les utilisateurs par services

Comprendre et configurer un protocole de routage dynamique

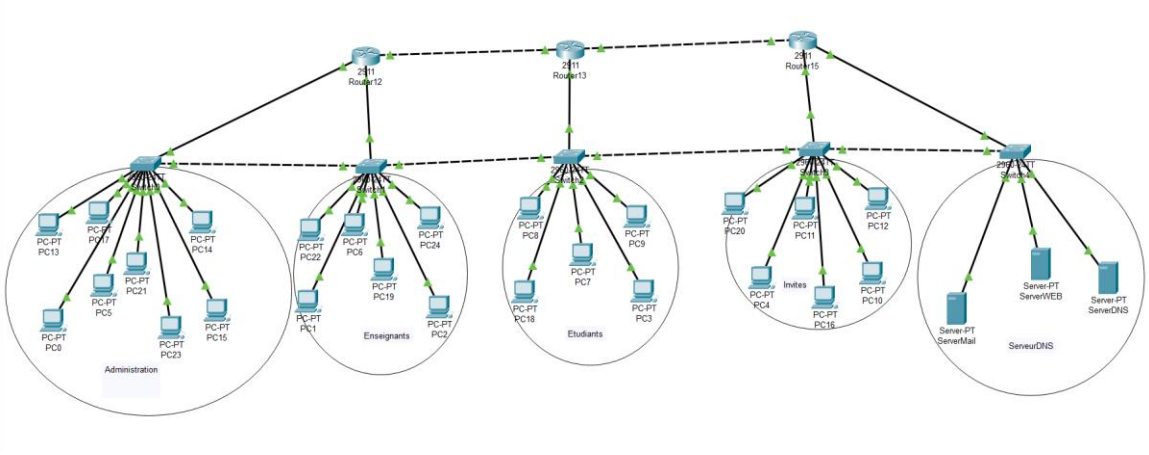
Tester la connectivité entre les différents réseaux

Création de la topologie réseau

3 Routeurs (R1, R2, R3)

5 Switches (S1 à S5)

25 PC



Configuration des switchs

Switch0

Physical Config CLI Attributes

IOS Command Line Interface

```
%LINK-5-CHANGED: Interface FastEthernet0/8, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/8, changed state to up


Switch>enable
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan10
                 ^
% Invalid input detected at '^' marker.

Switch(config)#vlan 10
Switch(config-vlan)#name Admin
Switch(config-vlan)#vlan 99
Switch(config-vlan)#name DNZ
Switch(config-vlan)#interface range fa0/1 -10
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 10
Switch(config-if-range)#interface fa0/24
Switch(config-if)#switchport acce
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (10), with Switch
FastEthernet0/1 (1).

% Incomplete command.
Switch(config-if)#schitchport access vlan 99
                 ^
% Invalid input detected at '^' marker.

Switch(config-if)#schwitchport access vlan 99
                 ^
% Invalid input detected at '^' marker.

Switch(config-if)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (10), with Switch
FastEthernet0/1 (1).

Switch(config-if)#switchport access vlan 99
Switch(config-if)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (10), with Switch
FastEthernet0/1 (1).
```

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Top

Switch2

Physical Config CLI Attributes

IOS Command Line Interface

```
%LINK-5-CHANGED: Interface FastEthernet0/6, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/6, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/7, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/7, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/8, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/8, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/9, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/9, changed state to up


Switch>enable
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 20
Switch(config-vlan)#name enseignants
Switch(config-vlan)#vlan 99
Switch(config-vlan)#name DNZ
Switch(config-vlan)#intewrface range fao/1 -20
                   ^
% Invalid input detected at '^' marker.

Switch(config-vlan)#interface range fa0/1 -20
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport  access vlan 20
Switch(config-if-range)#interface fa0/24
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (20), with Switch
FastEthernet0/2 (1).

Switch(config-if)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/2 (20), with Switch
FastEthernet0/1 (1).

Switch(config-if)#switchport  access vlan 99
Switch(config-if)#
```

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Switch5

Physical Config CLI Attributes

IOS Command Line Interface

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/6, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/7, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/7, changed state to up

%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/7 (1), with Switch
FastEthernet0/1 (20).

%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/7 (1), with Switch
FastEthernet0/1 (20).


Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 99
Switch(config-vlan)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/7 (1), with Switch
FastEthernet0/1 (20).

Switch(config-vlan)#name DNZ
Switch(config-vlan)#intewrface fa0/1 -20
                    ^
% Invalid input detected at '^' marker.

Switch(config-vlan)#intewrface range fa0/1 -20
                    ^
% Invalid input detected at '^' marker.

Switch(config-vlan)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/7 (1), with Switch
FastEthernet0/1 (20).
interface range fa0/1 -20
Switch(config-if-range)#interfa fa0/24
Switch(config-if)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/7 (1), with Switch
FastEthernet0/1 (20).

Switch(config-if)#switchport access vlan 99
Switch(config-if)#
```

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Delete

Switch3

Physical Config CLI Attributes

IOS Command Line Interface

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/11, changed state to up

%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (1), with Switch
FastEthernet0/2 (20).

%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (1), with Switch
FastEthernet0/2 (20).


Switch>enable
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 30
Switch(config-vlan)#name etudiants
Switch(config-vlan)#vlan 99
Switch(config-vlan)#name
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (1), with Switch
FastEthernet0/2 (20).

% Incomplete command.
Switch(config-vlan)#name  DNZ
Switch(config-vlan)#interface range fao/1 -30
                   ^
% Invalid input detected at '^' marker.

Switch(config-vlan)#interface range fa0/1 -30
interface range not validated - command rejected
Switch(config)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (1), with Switch
FastEthernet0/2 (20).
interface range fa0/1 -
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (1), with Switch
FastEthernet0/2 (20).

% Incomplete command.
Switch(config)#interface range fa0/1 -20
Switch(config-if-range)#switchport access vlan 30
Switch(config-if-range)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (30), with Switch
FastEthernet0/2 (20).
```

Delete

Configurations des routeur

Router12

Router12

Physical Config CLI Attributes

IOS Command Line Interface

```
         --- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]: no


Press RETURN to get started!



Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface g0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

Router(config-if)#interface g0/1
Router(config-if)#ip address 192.168.2.1 255.255.255.252
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router(config-if)#interface g0/2
Router(config-if)#ip address 192.168.3.1 255.255.255.0
Router(config-if)#shutdown
Router(config-if)#
Router(config-if)#
Router(config-if)#
Router(config-if)#
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up
```

PC13

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.30.0

Pinging 192.168.30.0 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.30.0:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>show ip route
Invalid Command.

C:\>ping 192.168.30.0

Pinging 192.168.30.0 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.30.0:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>show ip route
Invalid Command.

C:\>show ip interface brief
Invalid Command.
```

```
C:\>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:

Reply from 192.168.10.1: bytes=32 time=5ms TTL=128
Reply from 192.168.10.1: bytes=32 time=11ms TTL=128
Reply from 192.168.10.1: bytes=32 time=6ms TTL=128
Reply from 192.168.10.1: bytes=32 time=8ms TTL=128

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 5ms, Maximum = 11ms, Average = 7ms

C:\>
```

```
Router>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     192.168.7.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.7.0/30 is directly connected, GigabitEthernet0/1
L       192.168.7.1/32 is directly connected, GigabitEthernet0/1
     192.168.8.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.8.0/24 is directly connected, GigabitEthernet0/2
L       192.168.8.1/32 is directly connected, GigabitEthernet0/2

Router>
```

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to down


Router>show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.6.1     YES manual up                    down
GigabitEthernet0/1     192.168.7.1     YES manual up                    up
GigabitEthernet0/2     192.168.8.1     YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
Router>
```

Router12

Physical Config CLI Attributes

IOS Command Line Interface

```
Press RETURN to get started.

Router>
Router>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0
L       192.168.1.1/32 is directly connected, GigabitEthernet0/0
     192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.2.0/30 is directly connected, GigabitEthernet0/1
L       192.168.2.1/32 is directly connected, GigabitEthernet0/1
     192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.3.0/24 is directly connected, GigabitEthernet0/2
L       192.168.3.1/32 is directly connected, GigabitEthernet0/2

Router>
```

Router12

Physical Config CLI Attributes

IOS Command Line Interface

```
Router>
Router>
Router>show eigrg 100
             ^
% Invalid input detected at '^' marker.

Router>show ip eigrg neighbors
                   ^
% Invalid input detected at '^' marker.

Router>show ip eigrp neighbors
IP-EIGRP neighbors for process 100

Router>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0
L       192.168.1.1/32 is directly connected, GigabitEthernet0/0
     192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.2.0/30 is directly connected, GigabitEthernet0/1
L       192.168.2.1/32 is directly connected, GigabitEthernet0/1
     192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.3.0/24 is directly connected, GigabitEthernet0/2
L       192.168.3.1/32 is directly connected, GigabitEthernet0/2

Router>
```

```
Router>show ip eigrp neighbors
IP-EIGRP neighbors for process 100

Router>
```

Routeur13

Router13

Physical Config CLI Attributes

IOS Command Line Interface

```
Press RETURN to get started!



Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface g0/0
Router(config-if)#ip address 192.168.4.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#
Router(config-if)#interface g0/1
Router(config-if)#ip address 192.168.2.2 255.255.255.252
Router(config-if)#not shutdown
                  ^
% Invalid input detected at '^' marker.

Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

Router(config-if)#interface g0/2
Router(config-if)#ip address 192.168.5.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up

Router(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
```

Router13

Physical Config CLI Attributes

IOS Command Line Interface

```
Press RETURN to get started.

Router>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.2.0/30 is directly connected, GigabitEthernet0/1
L       192.168.2.2/32 is directly connected, GigabitEthernet0/1
     192.168.4.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.4.0/24 is directly connected, GigabitEthernet0/0
L       192.168.4.1/32 is directly connected, GigabitEthernet0/0
     192.168.5.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.5.0/24 is directly connected, GigabitEthernet0/2
L       192.168.5.1/32 is directly connected, GigabitEthernet0/2

Router>
```

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```
Router>show ip eigrp neighbors
IP-EIGRP neighbors for process 100

Router>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.2.0/30 is directly connected, GigabitEthernet0/1
L       192.168.2.2/32 is directly connected, GigabitEthernet0/1
     192.168.4.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.4.0/24 is directly connected, GigabitEthernet0/0
L       192.168.4.1/32 is directly connected, GigabitEthernet0/0
     192.168.5.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.5.0/24 is directly connected, GigabitEthernet0/2
L       192.168.5.1/32 is directly connected, GigabitEthernet0/2

Router>
```

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Routeur 15

```
L        192.168.6.1/32 is directly connected, GigabitEthernet0/2

Router>en
Router#conf
Configuring from terminal, memory, or network [terminal]? terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface g0/0
Router(config-if)#ip address 192.168.6.1
% Incomplete command.
Router(config-if)#ip address 192.168.6.1 255.255.255.0
Router(config-if)#
Router(config-if)#no shutdown
Router(config-if)#interface g0/1
Router(config-if)#ip address 192.168.7.1 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#interface g0/2
Router(config-if)#ip address 192.168.8.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#exit
                 ^
```

Router15

Physical Config CLI Attributes

IOS Command Line Interface

```
Would you like to enter the initial configuration dialog? [yes/no]: no


Press RETURN to get started!



Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface g0/0
Router(config-if)#ip address 192.168.6.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#interface g0/1
Router(config-if)#ip address 192.168.7.1 255.255.255.252
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router(config-if)#inteface g0/2
                  ^
% Invalid input detected at '^' marker.

Router(config-if)#interface g0/2
Router(config-if)#ip address 192.168.8.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up

Router(config-if)#
```

```
Router>show ip eigrp neighbors
IP-EIGRP neighbors for process 100

Router>
```

Router15

Physical Config CLI Attributes

IOS Command Line Interface

```
Router>show ip eigrp neighbors
IP-EIGRP neighbors for process 100

Router>show ip routeur
                     ^
% Invalid input detected at '^' marker.

Router>show ip router
                       ^
% Invalid input detected at '^' marker.

Router>enable
Router#show ip router
                       ^
% Invalid input detected at '^' marker.

Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     192.168.6.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.6.0/24 is directly connected, GigabitEthernet0/0
L       192.168.6.1/32 is directly connected, GigabitEthernet0/0
     192.168.7.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.7.0/30 is directly connected, GigabitEthernet0/1
L       192.168.7.1/32 is directly connected, GigabitEthernet0/1
     192.168.8.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.8.0/24 is directly connected, GigabitEthernet0/2
L       192.168.8.1/32 is directly connected, GigabitEthernet0/2

Router#
```

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Server Mail

Physical Config Services Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer SERVER Command Line 1.0
C:\>ping 192.168.2.10

Pinging 192.168.2.10 with 32 bytes of data:

Reply from 192.168.2.10: bytes=32 time=6ms TTL=128
Reply from 192.168.2.10: bytes=32 time=4ms TTL=128
Reply from 192.168.2.10: bytes=32 time<1ms TTL=128
Reply from 192.168.2.10: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.2.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 6ms, Average = 2ms

C:\>
```

ServerWEB

Physical Config Services Desktop Programming Attributes

Command Prompt

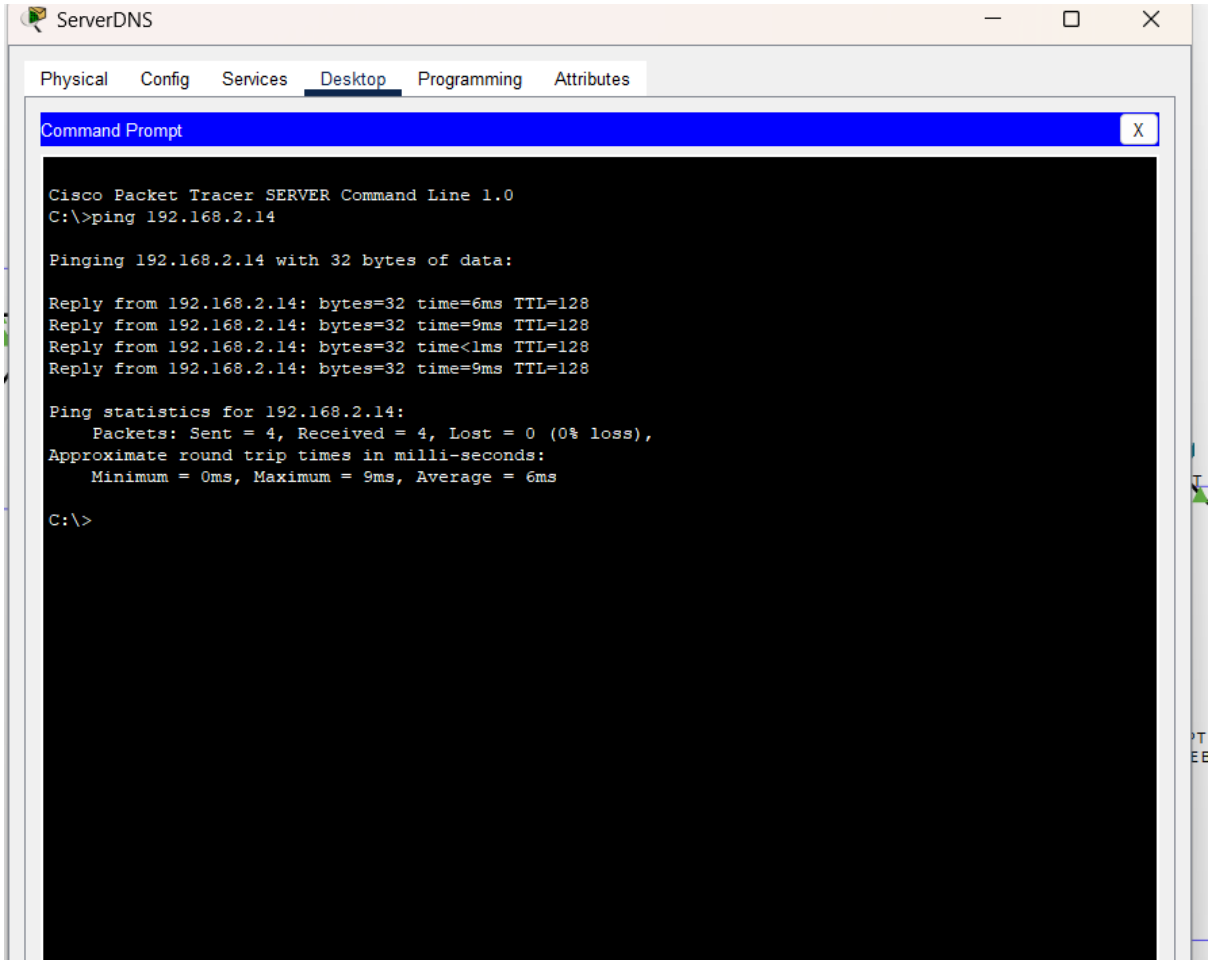
```
Cisco Packet Tracer SERVER Command Line 1.0
C:\>ping 192.168.2.13

Pinging 192.168.2.13 with 32 bytes of data:

Reply from 192.168.2.13: bytes=32 time=4ms TTL=128
Reply from 192.168.2.13: bytes=32 time=2ms TTL=128
Reply from 192.168.2.13: bytes=32 time=1ms TTL=128
Reply from 192.168.2.13: bytes=32 time=5ms TTL=128

Ping statistics for 192.168.2.13:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 5ms, Average = 3ms

C:\>
```



Questions TD

1. Pourquoi OSPF est plus efficace que RIP ?
   R) OSPF est plus efficace que le RIP parce qu'il choisit le chemin selon la bande passante réelle (pas juste le nombre de sauts), converge rapidement en cas de panne, n'envoie des mises à jour que si le réseau change, et supporte les grands réseaux sans limite de sauts.

2. Quelle différence entre protocole classful et classless ?
   R) Un protocole classful ex: RIP v1 n'envoie pas le masque de sous-réseau dans ses mises à jour il suppose des masques fixes (classe A, B, C). Un protocole classless ex: RIP v2, OSPF, EIGRP inclut le masque dans ses mises à jour → il supporte le

VLSM et le CIDR, ce qui permet une meilleure gestion des adresses IP.

3. Pourquoi EIGRP est plus rapide ?

R) EIGRP est plus rapide grâce à l'algorithme DUAL qui précalcule un chemin de secours Feasible Successor. En cas de panne, il bascule immédiatement sur ce chemin sans recalculer toute la table. Il envoie aussi des mises à jour partielles et déclenchées, pas périodiques.

4. Quel protocole choisir pour une grande entreprise et pourquoi ?

R) On choisit OSPF parce qu’il est idéal car il est ouvert compatible tous constructeurs, supporte les grands réseaux grâce aux aires, converge rapidement, et utilise une métrique basée sur la bande passante réelle. EIGRP est une alternative plus rapide mais propriétaire Cisco.